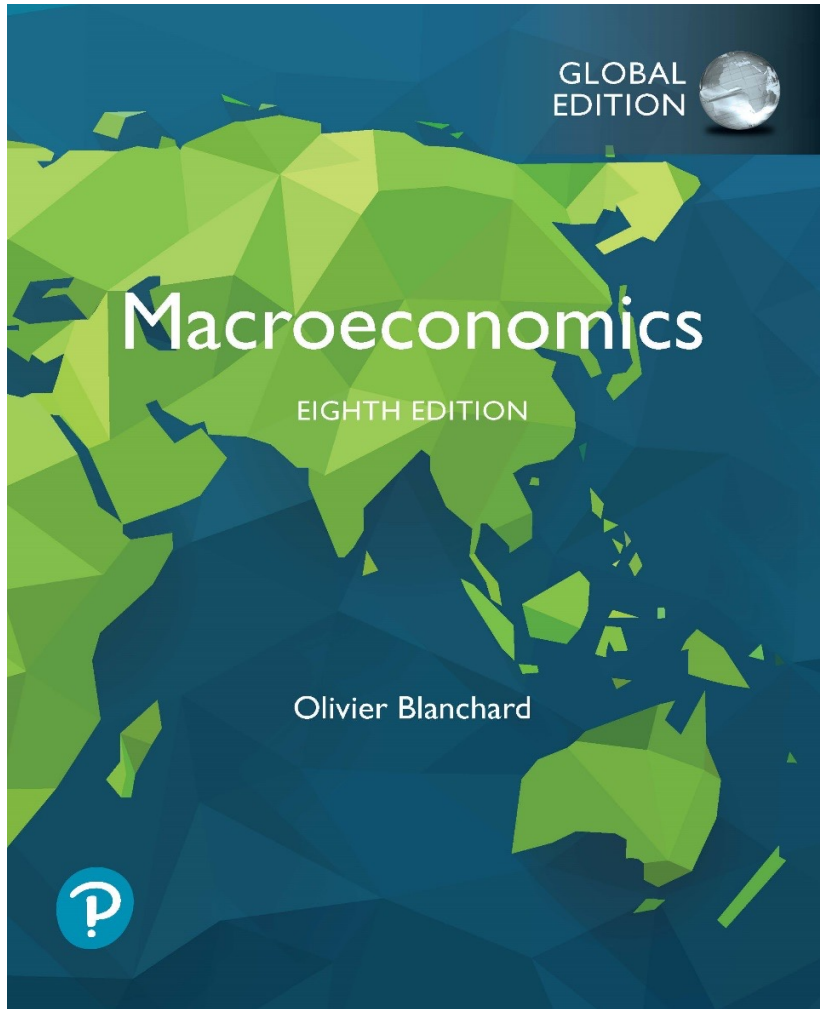


Macroeconomics

Eighth Edition, Global Edition



Chapter 10

The Facts of Growth

Chapter 10 Outline

The Facts of Growth

10.1 Measuring the Standard of Living

10.2 Growth in Rich Countries since 1950

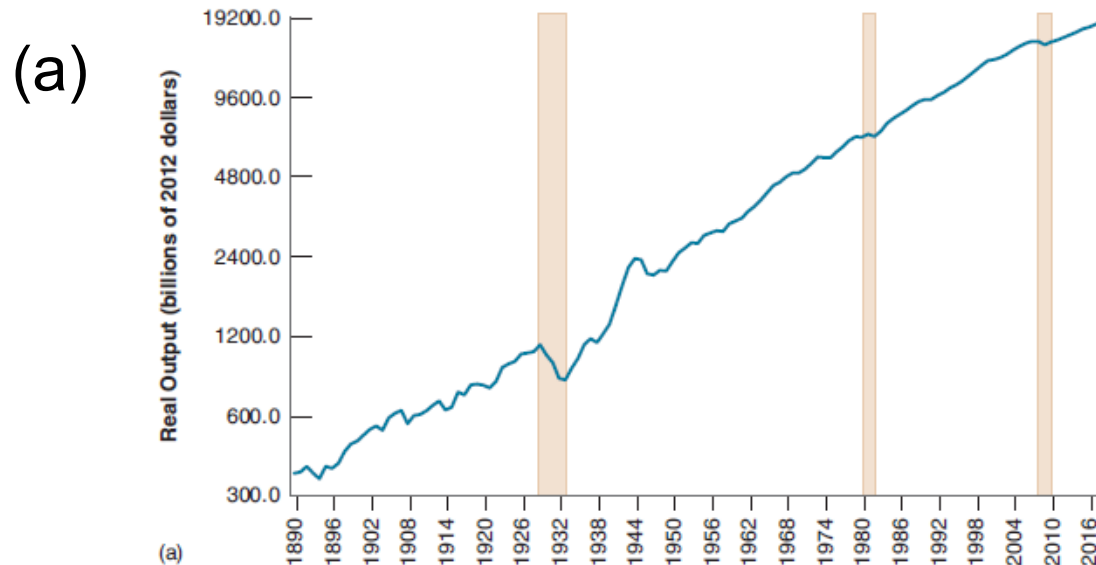
10.3 A Broader Look across Time and Space

10.4 Thinking about Growth: A Primer

The Facts of Growth

- **Growth** is the steady increase in aggregate output over time.
- We now shift our focus from economic fluctuations and the determination of output in the *short and medium run* to growth and the determination of output in the *long run*.

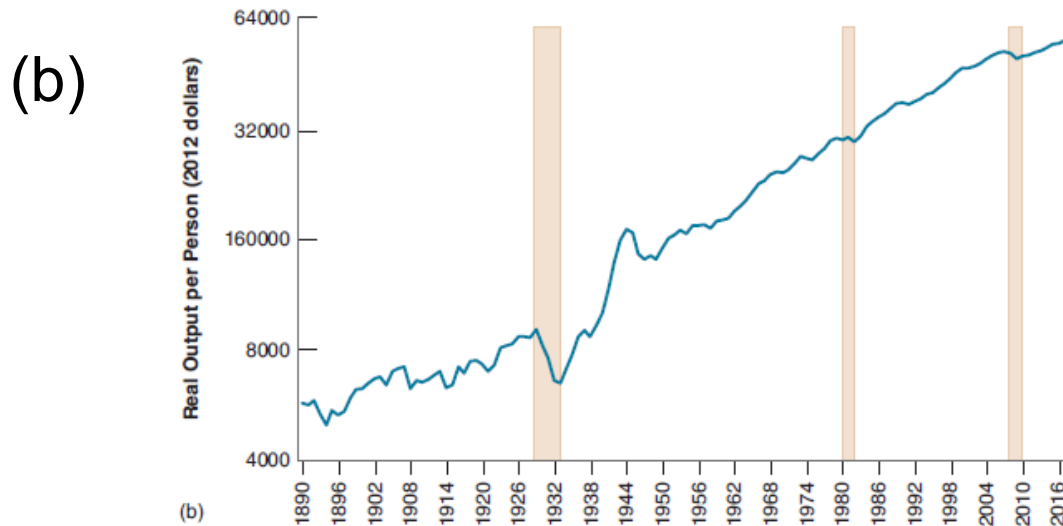
Figure 10.1 U.S. GDP since 1890 and U.S. GDP per person since 1890 (1 of 2)



Panel A shows the enormous increase in U.S. output since 1890, by a factor of 51.

- Source: 1890–1947: *Historical Statistics of the United States*. <http://hsus.cambridge.org/HSUSWeb/toc/hsusHome.do>. 1948–2014: *National Income and Product Accounts*. Population estimates 1890–2014: Louis Johnston and Samuel H. Williamson, “What Was the US GDP Then?” *Measuring Worth*, 2015, www.measuringworth.com/datasets/usgdp/. More recent data: FRED: GDPC1, B230RC0Q173SBEA.

Figure 10.1 U.S. GDP since 1890 and U.S. GDP per person since 1890 (2 of 2)



Panel *B* shows that the increase in output is not simply the result of the large increase in U.S. population from 63 million to more than 320 million over this period. Output per person has risen by a factor of 10.

- Source: 1890–1947: *Historical Statistics of the United States*. <http://hsus.cambridge.org/HSUSWeb/toc/hsusHome.do>. 1948–2014: *National Income and Product Accounts*. Population estimates 1890–2014: Louis Johnston and Samuel H. Williamson, “What Was the US GDP Then?” *Measuring Worth*, 2015, www.measuringworth.com/datasets/usgdp/. More recent data: FRED: GDPC1, B230RC0Q173SBEA.

10.1 Measuring the Standard of Living

- We care about growth because we care about the standard of living.
- Output per person, rather than output itself, is the variable we compare over time or across countries.
- We need to correct for variations in exchange rates and systematic differences in prices across countries.
- When comparing the standard of living across countries, we use **purchasing power parity (PPP)** numbers which adjust the numbers for the **purchasing power** of different countries.
- The right measure on the production side is *output per worker* or *output per hour worked*.

FOCUS: The Construction of PPP Numbers

- Consider this example:
 - United States: Each year people buy a new car for \$10,000 and spend another \$10,000 on food.
 - Russia: Each year people spend 40,000 rubles on cars (each lasts for 15 years) a year, and 80,000 rubles on food.
- If the exchange rate is $\$1 = 60$ rubles, consumption per person in Russia is only 10% of U.S. consumption per person.
- If we use U.S. prices for both countries and assume people spending all money on food, then consumption per person is \$20,000 ($\$10,000 + \$10,000$) in the U.S., but only \$10,700 in Russia, so Russian consumption per person is 53.5% of U.S. consumption per person.

10.2 Growth in Rich Countries since 1950 (1 of 2)

Table 10.1 The Evolution of Output per Person in Four Rich Countries since 1950

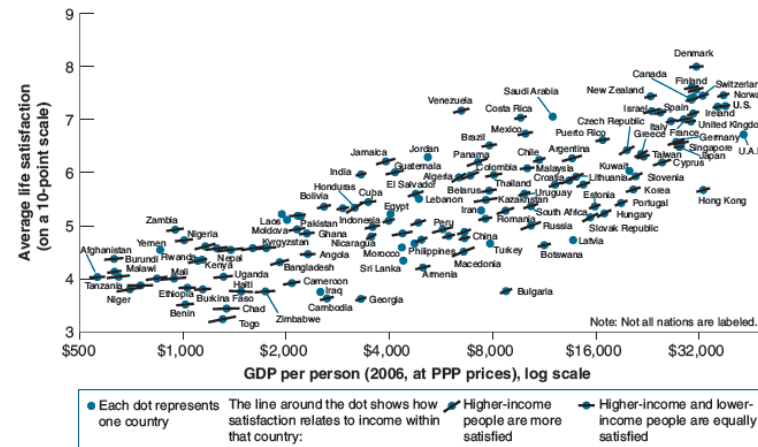
	Annual Growth Rate Output per Person (%)	Real Output per Person (2011 dollars)		
	1950–2017	1950	2017	2017/1950
France	2.6	7,025	39,461	5.6
Japan	4.1	2,531	40,374	15.9
United Kingdom	2.1	9,354	39,128	4.2
United States	2.0	14,569	54,995	3.8
Average	2.7	8,370	43,490	5.2

Notes: The data stop in 2017, the latest year (at this point) available in the Penn tables. The average in the last line is a simple unweighted average.

- *Source:* Penn World Table Version 8.1./Feenstra, Robert C., Robert Inklaar and Marcel P. Timmer (2015), “The Next Generation of the Penn World Table” forthcoming American Economic Review.

FOCUS: Does Money Lead to Happiness?

Figure 1 Life Satisfaction and Income per Person



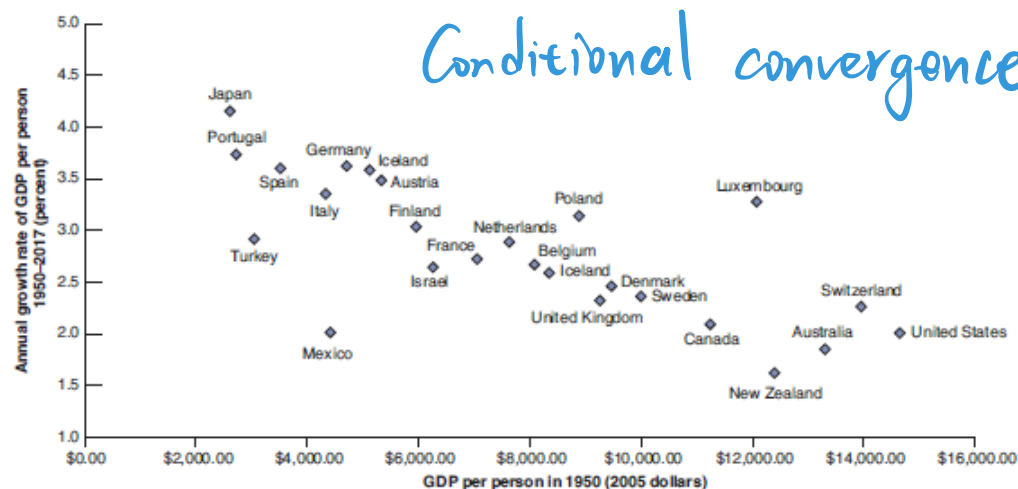
- **Easterlin paradox:** Once basic needs are satisfied, higher income per person does not increase happiness, and the level of income relative to others, rather than the absolute level of income, matters

Source: Betsey Stevenson and Justin Wolfers, Wharton School at the University of Pennsylvania. Used with Permission.

10.2 Growth in Rich Countries since 1950 (2 of 2)

Catch up for poor countries and convergen

Figure 10.2 Growth Rate of GDP Per Person since 1950 to rich versus GDP per Person in 1950; OECD Countries countries.



Countries with lower levels of output per person in 1950 have typically grown faster.

- Source: Penn World Tables Version 9. /Feenstra, Robert C., Robert Inklaar and Marcel P. Timmer (2015), "The Next Generation of the Penn World Table" American Economic Review, 105(10), 3150–3182, available for download at www.ggdc.net/pwt.

10.3 A Broader Look across Time and Space (1 of 5)

- From the end of the Roman Empire to roughly year 1500, Europe was in a **Malthusian trap** or *Malthusian era* with stagnation of output per person because most workers were in agriculture with little technological progress.
- After 1500, growth of output per person turned positive but still small.
- Between 1820 and 1950, U.S. growth was still 1.5% per year.
- Sustained growth was high since 1950.

10.3 A Broader Look across Time and Space (2 of 5)

Figure 10.3 Growth Rate of GDP per Person since 1960, versus GDP Per Person in 1960, 85 Countries

There is no clear relation between the growth rate of output since 1960 and the level of output per person in 1960.



- *Source:* Penn World Tables Version 9. /Feenstra, Robert C., Robert Inklaar and Marcel P. Timmer (2015), “The Next Generation of the Penn World Table” American Economic Review, 105(10), 3150–3182, available for download at www.ggdc.net/pwt.

10.3 A Broader Look across Time and Space (3 of 5)

- For the OECD countries, there is clear evidence of convergence.
- Convergence is also visible for many Asian countries, especially for those with high growth rates, called the four tigers—Singapore, Taiwan, Hong Kong, and South Korea.
- Most African countries were very poor in 1960, and some of them had negative growth of output per person between 1960 and 2017 due in part to internal or external conflicts.

10.3 A Broader Look across Time and Space (4 of 5)

Table 1 Annual workingman's Budget, Philadelphia, 1851, in current dollars

Item of expenditure	Amount (\$)	(% of total)
Butcher's meat (2 lb a day)	72.80	13.5
Flour (61/2 lb a year)	32.50	6.0
Butter (2 lb a week)	32.50	6.0
Potatoes (2 pk a week)	26.00	4.8
Sugar (4 lb a week)	16.64	3.0
Coffee and tea	13.00	2.4
Milk	7.28	1.3
Salt, pepper, vinegar, starch, soap, yeast, cheese, eggs	20.80	3.8
Total expenditures for food	221.52	41.1
Rent	156.00	28.9
Coal (3 tons a year)	15.00	2.7
Charcoal, chips, matches	5.00	0.9

10.3 A Broader Look across Time and Space (5 of 5)

Table 1 Annual workingman's Budget, Philadelphia, 1851, in current dollars
[continued]

Item of expenditure	Amount (\$)	(% of total)
Candles and oil	7.28	1.4
Household articles (wear, tear, and breakage)	13.00	2.4
Bedclothes and bedding	10.40	1.9
Wearing apparel	104.00	10.3
Newspapers	6.24	1.1
Total expenditures other than food	316.92	58.9

Sources: William Baumol, Sue Ann Batey Blackman, and Edward N. Wolff, *Productivity and American Leadership: The Long View* (1989, MIT Press), Chapter 3, Table 3.2. The composition of expenditures today comes from Appendix 9, Chapter 6, in the BLS *Handbook of Methods*.

10.4 Thinking About Growth: A Primer

(1 of 4)

- Aggregate production function F :

$$Y = F(K, N) \quad (10.1)$$

where Y is output, K is capital, and N is labor.

- The function F depends on the **state of technology**.
- **Constant returns to scale:**

$$xY = F(xK, xN) \quad (10.2)$$

- **Decreasing returns to capital:** Increases in capital lead to smaller and smaller increases in output.
- **Decreasing returns to labor:** Increases in labor lead to smaller and smaller increases in output.

10.4 Thinking About Growth: A Primer

(2 of 4)

- The production function and constant returns to scale imply a simple relation between output per worker (Y/N) and capital per worker (K/N):

$$\frac{Y}{N} = F\left(\frac{K}{N}, \frac{N}{N}\right) = F\left(\frac{K}{N}, 1\right) \quad (10.3)$$

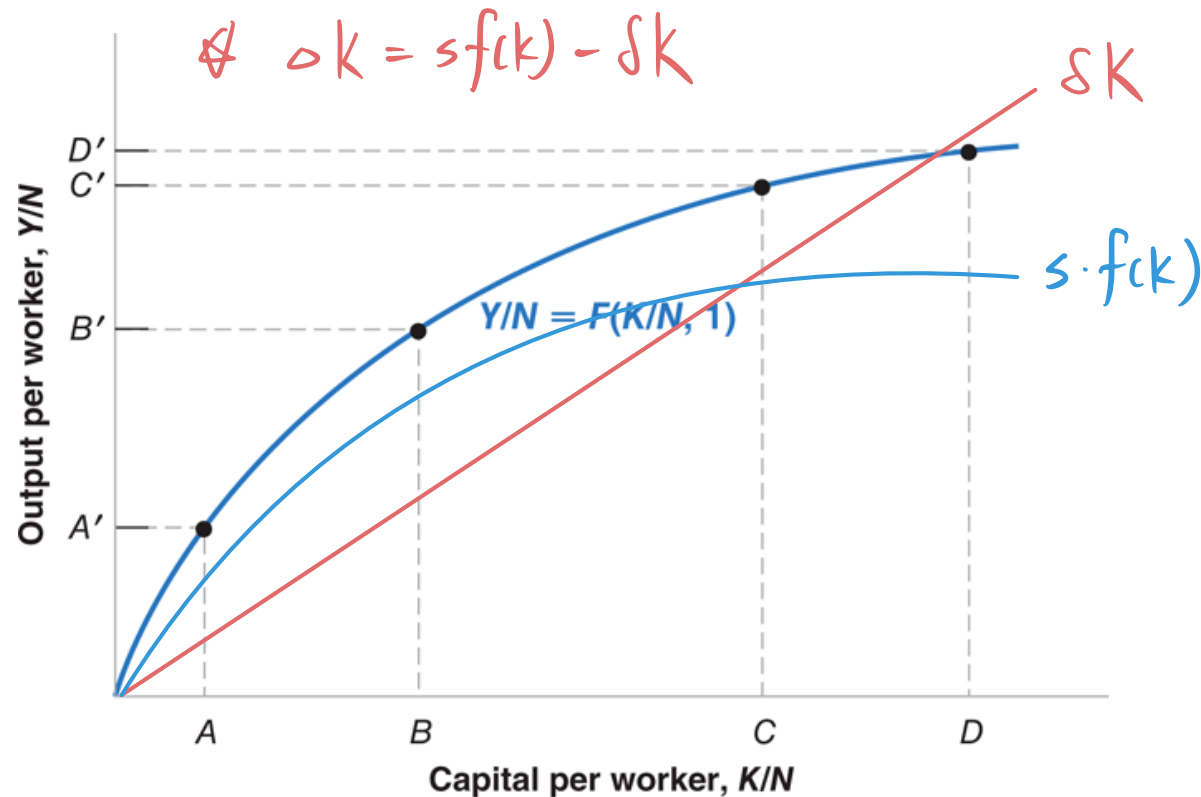
- Increases in capital per worker: Movements along the production function.
- Improvements in the state of technology: Shifts (up) of the production function.
- Growth comes from **capital accumulation** (a higher **saving rate**) and **technological progress** (the improvement in the state of technology).

10.4 Thinking About Growth: A Primer

(3 of 4)

Figure 10.4 Output and Capital per Worker

Decreasing returns to capital: Increases in capital per worker lead to smaller and smaller increases in output per worker.



10.4 Thinking About Growth: A Primer

(4 of 4)

Figure 10.5 The Effects of an Improvement in the State of Technology

An improvement in technology shifts the production function up, leading to an increase in output per worker for a given level of capital per worker.

